Project Report

CPU Data Analysis Report

Name :Mahnaz Alipour

Date:2026

Dataset:Ec2 CPU utilization (Kaggle)

Objective

Analyze CPU usage data from an EC2 server to understand normal patterns and detect unusual values.

Dataset information

Analyze CPU usage data from an EC2 server to understand normal patterns and detect unusual values.

Item	Values
Source	Kaggle
File name	Ec2 CPU utilization
Total Rows	4,033
Total Columns	2
Column name	Timestamp,value

Results

Metric	Value
Mean	0.1263
Maximum	2.344
Minimum	0.066

Anomaly Detection

Item	Values
Threshold	0.3
Anomalies Found	18
Percentage	0/45%

Findings

- Normal CPU usage is around12.63%
- 18 unusual spikes were detected
- Most anomalies occurred around 3:00am
- This suggests an automated daily task

Conclusion

The server normally operates at low CPU usage. The detected anomalies show a daily pattern that should be investigated to understand the cause of CPU spikes.

Code Summary

```
Import pandas as pd

# ==========================================
# CPU Data Analysis Project
# ==========================================

# File path
File_path = '/storage/emulated/0/Download/Python_project/ec2_cpu_utilization.csv'

# Read the file
Data = pd.read_csv(file_path)

# ==========================================
# Main Analysis
# ==========================================
Print("=" * 50)
Print("CPU DATA ANALYSIS REPORT")
Print("=" * 50)
```

```
# Show basic info
Print("\nTotal Rows:", len(data))
Print("Total Columns:", len(data.columns))
Print("Column Names:", list(data.columns))

# Show first 5 rows
Print("\n" + "-" * 50)
Print("First 5 Rows:")
Print("-" * 50)
Print(data.head())

# Show statistics
Print("\n" + "-" * 50)
Print("Statistics:")
Print("-" * 50)
Print("Mean:", round(data['value'].mean(), 4))
Print("Max:", round(data['value'].max(), 4))
Print("Min:", round(data['value'].min(), 4))

# Find anomalies
Print("\n" + "-" * 50)
Print("Anomaly Detection:")
Print("-" * 50)

Threshold = 0.3
Anomalies = data[data['value'] > threshold]

Print("Threshold:", threshold)
```

```
Print("Number of Anomalies:", len(anomalies))

If len(anomalies) > 0:
    Print("\nTop 5 Anomalies:")
    Print(anomalies.head())

# End
Print("\n" + "=" * 50)
Print("□ ANALYSIS COMPLETED!")
Print("=" * 50)
```